

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
```

```
df=pd.read_csv(r"C:\Users\Reddy Sai Abhiram\Downloads\trends.csv")
df
```

|       | location | year | category        | rank | query               |
|-------|----------|------|-----------------|------|---------------------|
| 0     | Global   | 2001 | Consumer Brands | 1    | Nokia               |
| 1     | Global   | 2001 | Consumer Brands | 2    | Sony                |
| 2     | Global   | 2001 | Consumer Brands | 3    | BMW                 |
| 3     | Global   | 2001 | Consumer Brands | 4    | Palm                |
| 4     | Global   | 2001 | Consumer Brands | 5    | Adobe               |
| ...   | ...      | ...  | ...             | ...  | ...                 |
| 26950 | Vietnam  | 2020 | Là Gì?          | 1    | Virus Corona là gì  |
| 26951 | Vietnam  | 2020 | Là Gì?          | 2    | Miền thị thực là gì |
| 26952 | Vietnam  | 2020 | Là Gì?          | 3    | Đầu cắt moi là gì   |
| 26953 | Vietnam  | 2020 | Là Gì?          | 4    | Bệnh bạch hầu là gì |
| 26954 | Vietnam  | 2020 | Là Gì?          | 5    | Đông Lào là gì      |

26955 rows × 5 columns

```
df.head()
```

|   | location | year | category        | rank | query |
|---|----------|------|-----------------|------|-------|
| 0 | Global   | 2001 | Consumer Brands | 1    | Nokia |
| 1 | Global   | 2001 | Consumer Brands | 2    | Sony  |
| 2 | Global   | 2001 | Consumer Brands | 3    | BMW   |
| 3 | Global   | 2001 | Consumer Brands | 4    | Palm  |
| 4 | Global   | 2001 | Consumer Brands | 5    | Adobe |

```
df.tail()
```

|       | location | year | category | rank | query               |
|-------|----------|------|----------|------|---------------------|
| 26950 | Vietnam  | 2020 | Là Gì?   | 1    | Virus Corona là gì  |
| 26951 | Vietnam  | 2020 | Là Gì?   | 2    | Miền thị thực là gì |
| 26952 | Vietnam  | 2020 | Là Gì?   | 3    | Đầu cắt moi là gì   |
| 26953 | Vietnam  | 2020 | Là Gì?   | 4    | Bệnh bạch hầu là gì |
| 26954 | Vietnam  | 2020 | Là Gì?   | 5    | Đông Lào là gì      |

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   location    26955 non-null  object
1   year        26955 non-null  int64
2   category    26955 non-null  object
3   rank        26955 non-null  int64
4   query       26955 non-null  object
dtypes: int64(2), object(3)
memory usage: 1.0+ MB
```

```
df.shape
```

(26955, 5)

df.describe()

|       | year         | rank         |
|-------|--------------|--------------|
| count | 26955.000000 | 26955.000000 |
| mean  | 2015.243369  | 3.000000     |
| std   | 3.564683     | 1.41424      |
| min   | 2001.000000  | 1.00000      |
| 25%   | 2013.000000  | 2.00000      |
| 50%   | 2016.000000  | 3.00000      |
| 75%   | 2018.000000  | 4.00000      |
| max   | 2020.000000  | 5.00000      |

df.columns

Index(['location', 'year', 'category', 'rank', 'query'], dtype='object')

df.dtypes

location object  
year int64  
category object  
rank int64  
query object  
dtype: object

df.isnull()

|       | location | year  | category | rank  | query |
|-------|----------|-------|----------|-------|-------|
| 0     | False    | False | False    | False | False |
| 1     | False    | False | False    | False | False |
| 2     | False    | False | False    | False | False |
| 3     | False    | False | False    | False | False |
| 4     | False    | False | False    | False | False |
| ...   | ...      | ...   | ...      | ...   | ...   |
| 26950 | False    | False | False    | False | False |
| 26951 | False    | False | False    | False | False |
| 26952 | False    | False | False    | False | False |
| 26953 | False    | False | False    | False | False |
| 26954 | False    | False | False    | False | False |

26955 rows × 5 columns

df.sum()

location GlobalGlobalGlobalGlobalGlobalGlobalGlobalGlob...  
year 54320885  
category Consumer BrandsConsumer BrandsConsumer BrandsC...  
rank 80865  
query NokiaSonyBMWPalmaAdobeNostradamusOsama bin Lade...  
dtype: object

df.duplicated()

0 False  
1 False  
2 False  
3 False  
4 False  
...  
26950 False  
26951 False  
26952 False  
26953 False  
26954 False

Length: 26955, dtype: bool

df.sum()

```
location    GlobalGlobalGlobalGlobalGlobalGlobalGlobalGlob...
year                54320885
category    Consumer BrandsConsumer BrandsConsumer BrandsC...
rank                80865
query       NokiaSonyBMWPalmAdobeNostradamusOsama bin Lade...
dtype: object
```

df.nunique()

```
location      83
year           20
category     2450
rank           5
query       18431
dtype: int64
```

Start coding or [generate](#) with AI.